

TROITSKOYE, Russian Federation

WMO: 316550

Lat: 49.450N

Lon: 136.567E

Elev: 37

StdP: 100.88

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.0	-29.3	-34.2	0.1	-31.1	-32.0	0.2	-28.6	10.4	-18.9	9.6	-19.1	1.0	180	0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	29.7	22.5	28.0	21.8	26.5	20.8	23.8	27.8	22.8	26.6	21.8	25.4	2.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	17.3	26.4	21.5	16.2	25.2	20.6	15.3	24.2	71.9	27.9	68.0	26.6	64.3	25.5	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.6	8.1	7.0	-35.5	32.1	3.4	1.6	-37.9	33.3	-39.9	34.3	-41.8	35.2	-44.2	36.4		
(5)			-35.6	25.6	3.4	1.4	-38.0	26.6	-40.0	27.4	-41.8	28.2	-44.3	29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.0	-20.9	-16.4	-7.5	3.8	12.2	18.1	21.8	20.0	13.8	4.7	-7.7	-18.7	(6)
(7)		DBStd	15.71	5.36	5.23	5.45	4.93	4.28	3.66	3.00	2.76	3.86	4.53	6.30	4.99	(7)
(8)		HDD10.0	4066	958	739	541	198	26	1	0	0	11	172	531	888	(8)
(9)		HDD18.3	6182	1217	972	799	438	193	49	7	15	142	421	781	1147	(9)
(10)		CDD10.0	1158	0	0	0	11	96	242	367	309	124	9	0	0	(10)
(11)		CDD18.3	231	0	0	0	1	4	41	116	66	5	0	0	0	(11)
(12)		CDH23.3	1413	0	0	0	3	52	283	759	297	18	0	0	0	(12)
(13)		CDH26.7	312	0	0	0	0	8	59	206	38	1	0	0	0	(13)
(14)	Wind	WSAvg	2.8	2.7	2.7	2.7	2.7	3.0	2.6	2.5	2.4	2.8	3.0	3.4	3.3	(14)
(15)	Precipitation	PrecAvg	645	15	17	25	38	61	66	101	116	77	44	34	27	(15)
(16)		PrecMax	864	42	64	62	99	128	130	219	198	166	96	92	91	(16)
(17)		PrecMin	486	2	1	0	5	18	12	0	55	25	6	4	4	(17)
(18)		PrecStd	112	9	13	19	21	30	32	56	43	36	19	20	20	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.4	0.0	7.8	22.8	28.0	30.6	32.6	29.4	25.3	20.1	8.7	-3.1	(19)
(20)			MCWB	-6.3	-2.7	3.1	12.9	17.8	22.7	23.1	23.8	19.6	13.6	5.0	-3.8	(20)
(21)		2%	DB	-8.9	-3.2	4.9	18.2	24.7	28.4	30.6	27.7	23.5	16.4	5.4	-7.1	(21)
(22)			MCWB	-9.8	-5.0	1.4	10.6	15.9	21.0	23.3	22.3	18.5	11.3	2.5	-7.9	(22)
(23)		5%	DB	-11.3	-5.9	2.5	14.7	22.5	26.6	29.0	26.3	21.6	13.5	3.3	-9.5	(23)
(24)			MCWB	-12.0	-7.4	-0.3	8.5	14.6	19.7	22.5	21.4	17.1	9.3	1.2	-10.4	(24)
(25)		10%	DB	-13.3	-8.2	0.7	11.7	20.4	24.7	27.3	25.0	19.8	11.4	1.1	-11.4	(25)
(26)			MCWB	-13.9	-9.3	-1.4	6.8	13.6	18.8	21.7	20.7	16.1	7.9	-0.7	-12.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.2	-2.8	3.7	13.9	19.6	24.4	25.3	24.9	20.6	14.5	5.5	-3.9	(27)
(28)			MCDWB	-5.6	-0.8	7.3	21.3	23.9	28.4	30.1	28.2	24.2	19.1	8.4	-3.0	(28)
(29)		2%	WB	-9.7	-5.0	1.6	11.2	17.1	22.1	24.2	23.2	19.1	11.9	2.8	-7.8	(29)
(30)			MCDWB	-9.1	-3.2	4.5	17.2	23.2	27.1	28.9	26.4	22.5	15.3	5.1	-7.1	(30)
(31)		5%	WB	-12.0	-7.4	0.1	9.2	15.5	20.6	23.3	22.2	17.9	10.1	1.3	-10.3	(31)
(32)			MCDWB	-11.3	-6.0	2.2	14.2	21.0	25.2	27.4	25.3	20.9	12.8	3.1	-9.6	(32)
(33)		10%	WB	-13.9	-9.4	-1.4	6.9	14.1	19.4	22.4	21.3	16.7	8.2	-0.4	-12.1	(33)
(34)			MCDWB	-13.4	-8.3	0.7	11.4	19.4	23.7	26.2	24.1	19.2	11.0	1.0	-11.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.7	9.2	9.6	9.5	10.0	9.1	8.3	7.6	8.8	8.3	6.8	6.6	(35)
(36)			MCDBR	7.8	10.0	10.0	13.3	14.0	11.8	10.9	9.7	10.7	11.3	8.0	7.6	(36)
(37)			MCWBR	7.1	8.6	7.4	7.7	7.7	5.8	5.1	5.1	6.2	7.3	5.9	7.0	(37)
(38)		5% WB	MCDBR	7.6	9.9	9.4	12.1	12.4	10.4	9.4	8.2	9.2	10.4	7.5	7.5	(38)
(39)			MCWBR	7.1	8.4	7.1	7.7	8.0	5.9	5.2	4.9	5.8	7.2	6.0	7.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.285	0.332	0.401	0.521	0.565	0.504	0.500	0.438	0.402	0.434	0.370	0.295	(40)	
(41)		taud	2.075	1.949	1.838	1.766	1.769	2.024	2.080	2.266	2.280	1.996	2.000	2.054	(41)	
(42)		Ebn at Noon	732	774	777	723	712	768	764	799	788	649	599	650	(42)	
(43)		Edh at Noon	86	126	169	204	212	165	154	123	109	121	94	77	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.70	2.91	4.37	4.78	5.26	5.74	5.33	4.58	3.60	2.29	1.42	1.26	(44)	
(45)		RadStd	0.07	0.09	0.23	0.47	0.42	0.64	0.45	0.30	0.25	0.17	0.14	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		+0.26	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page